

STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2004

MATHEMATICS EXTENSION 2

Time Allowed - Three hours

GENERAL INSTRUCTIONS:

- Reading time - 5 minutes.
- Working time - 3 hours.
- Write using blue or black pen.
- Start each question on a new page.
- Write your Student Number at the top of each page.
- Calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every Question.

QUESTION 1

- (a) Use the substitution $u = x^2$ to calculate.

$$\int_0^{\frac{1}{\sqrt{2}}} \frac{x dx}{\sqrt{1-x^4}}.$$

- (b) Find: $\int \sec^4 x \, dx$.

- (c) By rationalising the numerator, show that:

$$\text{Show that } \int_0^1 \sqrt{\frac{1+x}{3+x}} \, dx = \sqrt{8} - \sqrt{3} + \ln \left(\frac{2+\sqrt{3}}{3+\sqrt{8}} \right).$$

- (d) Find real numbers A, B and C such that

$$\frac{4x+2}{(x+3)(x^2+1)} \equiv \frac{A}{x+3} + \frac{Bx+C}{x^2+1} \quad \text{and hence find:}$$

$$\int \frac{4x+2}{(x+3)(x^2+1)} \, dx$$

- (e) Use integration by parts to find $\int x \tan^{-1} x \, dx$

QUESTION 2.

(a) let $z = 1 + 2i$ and $\omega = 2 - i$. Find in the form $x + iy$.

(i) $z \bar{\omega}$.

(ii) $\frac{1}{\omega^2}$.

(b) Sketch the region in the complex plane where the inequalities

$$|z - 3 - 4i| \leq 5 \quad \text{and} \quad 0 \leq \arg z \leq \frac{\pi}{4} \quad \text{both hold.}$$

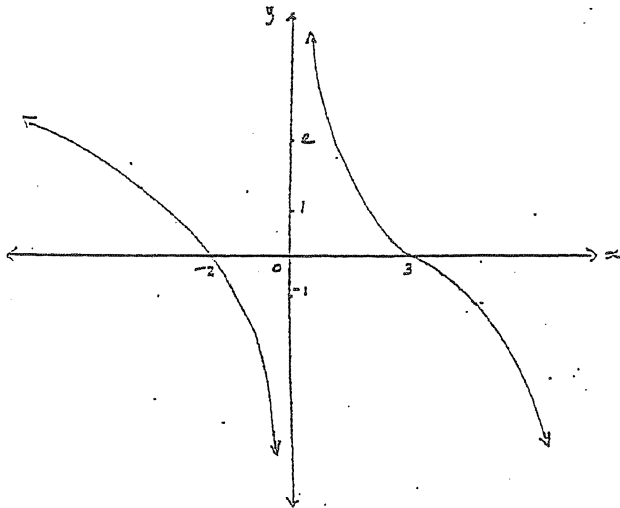
(c) Given that $1 - i$ is a root of $x^3 - 3x^2 + 4x - 2 = 0$, find the other two roots.

(d) If $\sqrt{3} + i$ and two other complex numbers form the vertices of an equilateral triangle on the complex plane with its centre at the origin, find the two other complex numbers.

(e) Express $(-1 + i)$ in modulus argument form and hence evaluate $(-1 + i)^8$.

QUESTION 3

- (a) The diagram shows the graph of $y = f(x)$.



Draw separate half page sketches of the graphs of the following showing all obvious features.

(i) $y = \frac{1}{f(x)}$

(ii) $y = \sqrt{f(x)}$

(iii) $y = f(|x|)$

(iv) $y = \ln f(x)$

- (b) An ellipse centred at the origin has a focus at $(-4, 0)$ and a directrix $x = 9$, find its equation.

- (c) The curve $y = \frac{ax^2 + bx + c}{cx^2 + bx + a}$ has a horizontal asymptote of $y = 4$. Find where this curve cuts the y axis.

- (d) If z lies on the locus of $|z - 2| = 2$ in the Argand diagram, show that:

$|z|^2 + |z - 4|^2$ is a constant

QUESTION 4

(a) The area bounded by the curve $y = 12x - x^2$, the x axis, $x = 2$ and $x = 10$ is rotated about the y axis to form a solid. By using the method of cylindrical shells calculate the volume of the solid.

(b) The base of a solid is the segment of the parabola $x^2 = 4y$ cut off by the line $y = 2$. Each cross section perpendicular to the y axis is a right angled isosceles triangle with hypotenuse in the base of the solid. Find the volume of the solid.

(c) The normal at a point $P(cp, \frac{c}{p})$ on the rectangular hyperbola $xy = c^2$ meets the x axis at Q. M is the mid point of PQ.

(i) By proving that the equation of this normal is:

$$y - \frac{c}{p} = p^2(x - cp)$$

show that M is the point $\left(\frac{2cp^4 - c}{2p^3}, \frac{c}{2p} \right)$.

(ii) Hence show that the locus of M is given by:

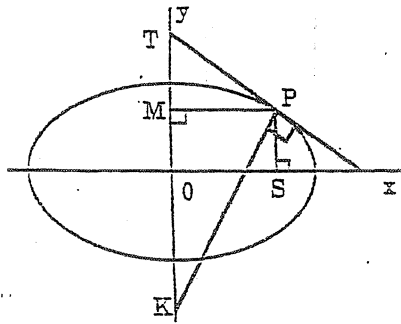
$$2xy c^2 = c^4 - 8y^4.$$

QUESTION 5

- (a) The diagram shows the point $P(a \cos \theta, b \sin \theta)$ on the ellipse.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

P lies vertically above S, one of the foci of the ellipse. The tangent at P meets the y-axis at T and the normal at P meets the y-axis at K. M is the foot of the perpendicular from P to the y-axis.



- (i) Write down the co-ordinates of S.
- (ii) Show that $\sin \theta = \sqrt{1 - e^2}$ where e is the eccentricity of the ellipse
- (iii) Find the co-ordinates of T, M and K. You may assume that the equation of the tangent at P is : $y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$
- (iv) Prove that the area of triangle TPK is:

$$A = \frac{abe}{2} \times \frac{(e^2 + 1)}{\sqrt{1 - e^2}}$$

- (b) The cubic $y = x^3 - px + q$ has two turning points

- (i) Show that $p > 0$.
- (ii) Find the co-ordinates of these turning points.
- (iii) The line $y = k$ meets the cubic in 3 points.

Show that $q - \frac{2p}{3} \sqrt{\frac{p}{3}} < k < q + \frac{2p}{3} \sqrt{\frac{p}{3}}$

- (c) Sketch without using calculus

$$y = \frac{2(x^2 - 8x)}{x^2 + x - 20}$$

QUESTION 6

- (a) When $p(x) = x^4 + ax^2 + bx$ is divided by $x^2 + 1$ the remainder is $x + 2$. Find the values of a and b .

- (b) Given that the equation $ax^4 + 4bx + c = 0$ has a double root, prove that

$$ac^3 = 27b^4.$$

- (c) The cubic equation $x^3 + px + q = 0$ has 3 real non zero roots α, β, χ . Find in terms of the constants p and q the value of:

(i) $\alpha^2 + \beta^2 + \chi^2$.

(ii) $(\alpha - 1)(\beta - 1)(\chi - 1)$.

(iii) $\alpha^3 + \beta^3 + \chi^3$.

(iv) $\alpha^4 + \beta^4 + \chi^4$.

QUESTION 7

- (a) A particle of mass m is projected downwards under gravity, in a medium whose resistance is equal to the velocity of the particle multiplied by $\frac{mg}{T}$. Show that the terminal velocity of this particle is T .

If a particle is projected vertically upwards in the same medium with velocity u , show that it attains a height of:

$$H = \frac{uT}{g} + \frac{T^2}{g} \log\left(\frac{T}{u+T}\right).$$

- (b) For what rational value (s) of k do the two equations:

$$2x^2 - 7x + k = 0 \quad \text{and}$$

$$2x^3 - x^2 - 37x + 36 = 0,$$

have a common root?

- (c) Show that $(1+x)^n (1 + \frac{1}{x})^n = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^{2n}$ and hence prove that

$$\sum_{r=0}^n \binom{n}{r}^2 = {}^{2n}C_n.$$

QUESTION 8

(a) By using De Moivre's Theorem prove that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad \text{and hence solve the equation:}$$

$$16x^5 - 20x^3 + 5x = 0$$

and deduce the values of

(i) $\cos \frac{\pi}{10}$.

(ii) $\cos \frac{3\pi}{10}$.

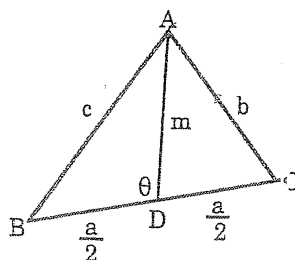
(b) Given that $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$

Prove that $I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1)I_{n-2}$ and hence evaluate:

$$\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$$

(c) In $\triangle ABC$ a median is drawn from A to meet BC at D. Prove that the length of the median is given by:

$$m = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$$



STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

1a) $\int_0^{\frac{1}{\sqrt{2}}} \frac{x dx}{\sqrt{1-x^2}}$
 Let $u = x^2$ $x = \frac{1}{\sqrt{2}}$ $u = \frac{1}{2}$
 $du = 2x dx$ $x=0$ $u=0$
 $\therefore I = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{du}{\sqrt{1-u}}$
 $= \frac{1}{2} [2\sqrt{1-u}]_0^{\frac{1}{2}}$
 $= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right]$
 $= \frac{\pi}{12}$ (3)

b) $\int \sec^4 x dx$
 $= \int (1 + \tan^2 x) \sec^2 x dx$
 $= \int \sec^2 x dx + \int \tan^2 x \sec^2 x dx$
 $= \tan x + \frac{1}{3} \tan^3 x + C$ (2)

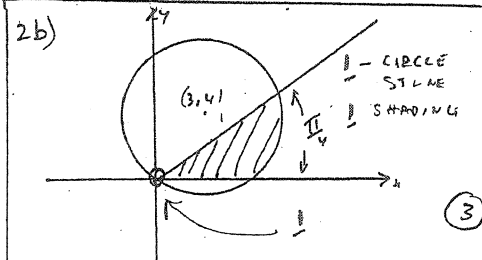
c) $\int_0^1 \sqrt{\frac{1+x}{3+x}} \times \frac{1}{\sqrt{1+x}} dx$
 $= \int_0^1 \frac{1+x}{\sqrt{3+4x+x^2}} dx$
 $= \int_0^1 \frac{\frac{1}{2}(2x+4) - 1}{\sqrt{(x+1)^2 - 1}} dx$
 $= \left[\sqrt{x^2+4x+3} \right]_0^1 - \left[\ln(x+2+\sqrt{x^2+4x+3}) \right]_0^1$
 $= \sqrt{8} - \sqrt{3} - \left[\ln(3+\sqrt{8}) - \ln(2+\sqrt{3}) \right]$
 $= \sqrt{8} - \sqrt{3} + \ln(2+\sqrt{3}) - \ln(3+\sqrt{8})$
 $= \sqrt{8} - \sqrt{3} + \ln\left(\frac{2+\sqrt{3}}{3+\sqrt{8}}\right)$ (5)

d) $A(x^2+1) + (x+3)(Bx+C) \equiv 4x+2$
 Let $x = -3$
 $\therefore 10A = -10 \Rightarrow A = -1$
 Let $x = 0$ $A + 3C = 2$
 $\therefore 3C = 2 \Rightarrow C = \frac{2}{3}$
 Then in x^2 $A + B = 0$ $A = -1$
 $\therefore B = 1$ $B = 1$
 $C = \frac{2}{3}$ (3)
 $\therefore \frac{4x+2}{(x+3)(x^2+1)} = \frac{-1}{x^2+1} + \frac{1}{x+3} - \frac{\frac{2}{3}}{x^2+1}$
 $= \frac{1}{2} \int \frac{2x dx}{x^2+1} + \int \frac{dx}{x+3} - \frac{2}{3} \int \frac{dx}{x^2+1}$
 $= \frac{1}{2} \ln(x^2+1) + \ln|x+3| - \frac{2}{3} \arctan x + C$ (4)

e) $I = \int x \tan^{-1} x dx$
 Let $u = \tan^{-1} x$ $v' = x$
 $u' = \frac{1}{1+x^2}$ $v = \frac{x^2}{2}$
 $\therefore I = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2+1-1}{1+x^2} dx$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{1}{2} \tan^{-1} x + C$ (3)
 $\boxed{17}$

(2) i) $z \bar{w} = (1+2i)(2+i)$
 $= (2-2) + (4+1)i$
 $= 5i$

ii) $\frac{1}{w^2} = \frac{1}{(2-i)^2}$
 $= \frac{1}{3-4i} \times \frac{3+4i}{3+4i}$
 $= \frac{3+4i}{25}$
 $= \frac{3}{25} + \frac{4}{25}i$ (4)

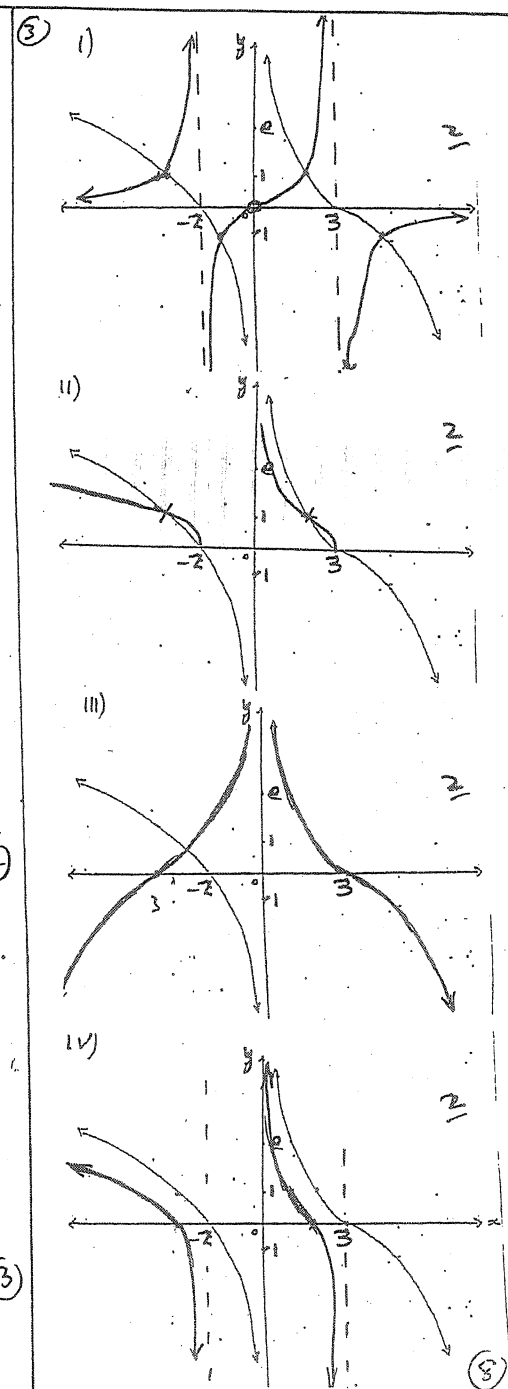


c) if $(1-i)$ is a root so is $(1+i)$
 $\therefore P(x) = (x-1)(x-(1-i))(x-(1+i))$
 $= (x-1)(x^2-2x+2)$
 $-2d = -2$
 $\therefore d = +1$
 $\therefore$ other two roots are $(1+i)$ and 1 (3)

d) if $z_1 = \sqrt{3} + i$
 $z_2 = (\sqrt{3} + i) \cos \frac{2\pi}{3}$
 $= \left(\frac{\sqrt{3}}{2} - \frac{i}{2} \right) + i \left(\frac{3}{2} - \frac{1}{2} \right)$
 $= -\sqrt{3} + i$
 $z_3 = (\sqrt{3} + i) \cos -\frac{2\pi}{3}$
 $= \left(-\frac{\sqrt{3}}{2} + \frac{i}{2} \right) + i \left(-\frac{1}{2} - \frac{3}{2} \right)$
 $= 0 + 2i$ (2)

$\therefore$ other 2 complex nos are $- \sqrt{3}, 2i$

e) $(-1+i)^8 = (\sqrt{2})^8 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)^8$
 $= \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$
 $\text{or } = \sqrt{2} \cos \frac{3\pi}{4}$
 $(-1+i)^8 = (\sqrt{2})^8 \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}$
 $= 16 (\cos 0 + i \sin 0)$
 $= 16$ (3)
 $\boxed{15}$



3b)

$$ae = 4$$

$$\frac{a}{e} = 9$$

$$\therefore a^2 = 36 \quad \downarrow$$

$$a = 6 \quad e = \frac{2}{3} \quad \downarrow$$

$$b^2 = a^2(1 - e^2)$$

$$b^2 = 36(1 - \frac{4}{9})$$

$$= 36 \cdot \frac{5}{9}$$

$$b = \sqrt{20} \quad \downarrow$$

$$\therefore \text{ellipse is } \frac{x^2}{36} + \frac{y^2}{20} = 1$$

(3)

$$c) \text{ as } x \rightarrow \infty$$

$$y \rightarrow \frac{a}{e} = 4 \quad \downarrow$$

cuts y axis when $x = 0$

$$\text{i.e. } y = \frac{c}{e}$$

$$= \frac{1}{\frac{1}{4}}$$

(2)

$$d) \text{ let } z = x + iy$$

$$\therefore |z - 2|^2 = (x - 2)^2 + y^2 = 4$$

$$\therefore x^2 - 4x + 4 + y^2 = 0 \quad \downarrow$$

$$|z|^2 + |z - 4|^2 = x^2 + y^2 + (x - 4)^2 + y^2 + 16$$

$$= x^2 + y^2 + x^2 - 8x + 16 + y^2 + 16$$

$$= 2(x^2 - 4x + y^2) + 16$$

$$= 16$$

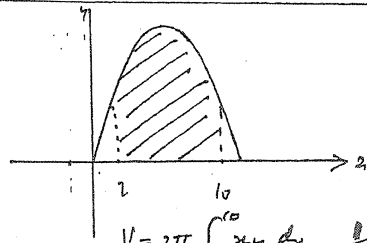
(2)

$$\therefore |z|^2 + |z - 4|^2 = 16 \quad \text{i.e. a}$$

circle

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4a)



$$V = 2\pi \int_2^{10} xy \, dx \quad \downarrow$$

$$= 2\pi \int_2^{10} x(12x - x^2) \, dx \quad \downarrow$$

$$= 2\pi \int_2^{10} (12x^2 - x^3) \, dx$$

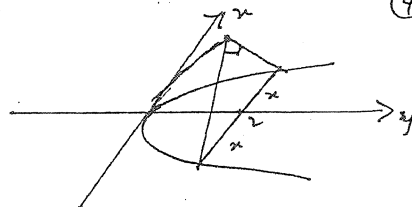
$$= 2\pi \left[4x^3 - \frac{x^4}{4} \right]_2^{10} \quad \downarrow$$

$$= 2\pi [4000 - 2500] - 2\pi [32 - 4]$$

$$= 2944\pi \text{ sq units} \quad \downarrow$$

(4)

b)



$$\text{Area } \triangle = \frac{1}{2} \times x \times x \times \sin \theta$$

$$= x^2 \sin \theta$$

$$S.V. = x^2 \sin \theta$$

$$\therefore V = \int_0^2 x^2 \sin \theta \, dx$$

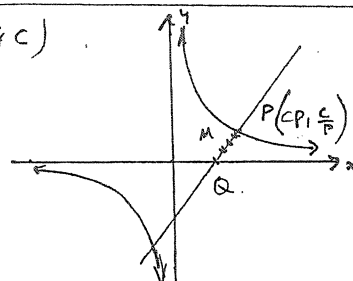
$$= \int_0^2 4y \, dy \quad \downarrow$$

$$= \left[2y^2 \right]_0^2$$

$$= 8 \text{ sq units} \quad \downarrow$$

(3)

4c)



1)

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$y' = -\frac{c^2}{x^2}$$

$$\text{substit } x = cp \quad \therefore y' = -\frac{1}{p^2}$$

$$\therefore \text{gradient of normal} = p^2$$

$$\text{equation of N is } y - \frac{c}{p} = p^2(x - cp) \quad \downarrow$$

$$\text{substit } y = 0 \text{ find Q}$$

$$\therefore -\frac{c}{p} = p^2(x - cp)$$

$$x - cp = -\frac{c}{p^3}$$

$$x = \left(cp - \frac{c}{p^3} \right)$$

$$\therefore \text{Q is at } \left(cp - \frac{c}{p^3}, 0 \right) \quad \downarrow$$

$$\text{M is at } \left(\frac{(cp - \frac{c}{p^3}) + cp}{2}, \frac{c}{2p} \right)$$

$$= \left(\frac{2cp^4 - c}{2p^3}, \frac{c}{2p} \right) \quad \downarrow$$

$$\text{ii) } 2xy^2 = 2 \left(\frac{2cp^4 - c}{2p^3} \right) \cdot \frac{c}{2p} \cdot c^2$$

$$= \frac{2c^4p^4 - c^4}{2p^4} \quad \downarrow$$

$$c^4 - 8y^4 = c^4 - 8 \times \left(\frac{c}{2p} \right)^4$$

$$= c^4 - \frac{8c^4}{16p^4}$$

$$= \frac{2c^4p^4 - c^4}{2p^4} \quad \downarrow$$

(6)

$$\therefore 2xy^2 = c^4 - 8y^4$$

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$$\text{5) a) i) S is at } (a \cos \theta, 0) \text{ or } (a, 0) \quad \downarrow$$

$$\text{ii) } a \cos \theta = a \cos \theta$$

$$r = a \cos \theta$$

$$a \cos \theta = \frac{1 - \cos \theta}{1 - e}$$

$$= \frac{1 - \cos \theta}{1 - e}$$

$$\text{iii) to find T substit } x = 0$$

in Tangent equation.

$$y - b \sin \theta = \frac{-b \cos \theta}{e \sin \theta} \cdot -e \cos \theta$$

$$\therefore y = \frac{b \cos \theta}{\sin \theta} + b \sin \theta$$

$$= \frac{b \cos^2 \theta + b \sin^2 \theta}{\sin \theta}$$

$$= \frac{b}{\sin \theta}$$

$$= \frac{b}{\sin \theta}$$

$$\text{T is point } \left(0, \frac{b}{\sin \theta} \right) \quad \downarrow$$

$$\text{to find K substit } x = 0$$

into equation of normal

$$y - b \sin \theta = \frac{a \cos \theta}{b \cos \theta} (x - a \cos \theta)$$

$$\therefore y = b \sin \theta - \frac{a^2 \sin \theta}{b}$$

$$= \frac{b^2 \sin \theta - a^2 \sin \theta}{b}$$

$$= \sin \theta \frac{b^2 - a^2}{b} \quad \downarrow$$

$$\therefore \text{K is at } \left(0, \frac{\sin \theta (b^2 - a^2)}{b} \right)$$

(5) i)

(i) S in pt $(a \cos \theta, 0)$ or $(ae, 0)$

(ii) $ae = a \cos \theta$
 $e = \cos \theta$
 $\sin \theta = \sqrt{1 - \cos^2 \theta}$
 $= \sqrt{1 - e^2}$

(iii) to find T in pt. $x = 0$
 in tangent eqn
 $\therefore y - b \cos \theta = \frac{-b \sin \theta}{a \sin \theta} x - a \sin \theta$
 $y = \frac{b \cos^2 \theta}{\sin \theta} + b \sin \theta$
 $= \frac{b(\cos^2 \theta + \sin^2 \theta)}{\sin \theta}$
 $= \frac{b}{\sin \theta}$

T in pt $(0, \frac{b}{\sin \theta})$ *

to find K in pt $x = 0$ into
 equation of the circle
 $y - b \cos \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$
 $\therefore y = b \sin \theta - \frac{a \sin \theta \cdot a \cos \theta}{b \cos \theta}$
 $= b \sin \theta - \frac{a^2 \sin \theta}{b}$
 $= \frac{b^2 \sin \theta - a^2 \sin \theta}{b}$
 $= \sin \theta \frac{(b^2 - a^2)}{b}$

$\therefore$ K in pt $(0, \frac{\sin \theta (b^2 - a^2)}{b})$ *

Min pt $(0, b \sin \theta)$ *

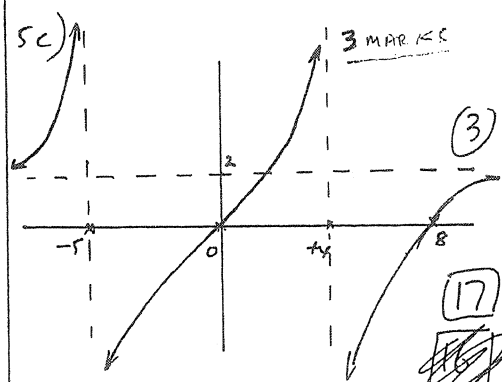
iv) Area $\triangle TPK$
 $= \frac{1}{2} \times TK \times MP$
 $= \frac{1}{2} \times \left(\frac{b}{\sin \theta} - \frac{(b^2 - a^2)}{b} \right) \times a \cos \theta$
 $= \frac{1}{2} \left(\frac{b^2 + a^2 \sin^2 \theta - b^2 \sin^2 \theta}{b \sin \theta} \right) a \cos \theta$
 $= \frac{(b^2 + a^2(1 - e^2) - b^2(1 - e^2)) \cdot ae}{2b \sqrt{1 - e^2}}$
 $= \frac{(b^2 + a^2 - a^2 e^2 - b^2 + b^2 e^2) ae}{2b \sqrt{1 - e^2}}$
 $= \frac{(a^2(1 - e^2) + e^2 b^2) ae}{2b \sqrt{1 - e^2}}$

Just as $b^2 = a^2(1 - e^2)$
 $\therefore$ Area $= \frac{(b^2 + e^2 b^2) ae}{2b \sqrt{1 - e^2}}$
 $= \frac{b^2(e^2 + 1) \cdot ae}{2b \sqrt{1 - e^2}}$
 $= \frac{abe(e^2 + 1)}{2\sqrt{1 - e^2}}$ (10)

b) $y = x^3 - px + q$
 turning pts where $y' = 0$
 i.e. $3x^2 - p = 0$
 to have 2 turning pts
 $3x^2 - p = 0$ has 2 roots
 i.e. $x^2 = \pm \sqrt{\frac{p}{3}}$
 $\therefore p > 0$ for 2 roots.

when $x = +\sqrt{\frac{p}{3}}$
 $y = \frac{p}{3} \sqrt{\frac{p}{3}} - p \sqrt{\frac{p}{3}} + q$

$= q - \frac{2p}{3} \sqrt{\frac{p}{3}}$
 when $x = -\sqrt{\frac{p}{3}}$
 $y = -\frac{p}{3} \sqrt{\frac{p}{3}} - p \sqrt{\frac{p}{3}} + q$
 $= q - \frac{4p}{3} \sqrt{\frac{p}{3}}$
 as $p > 0$
 $q - \frac{2p}{3} \sqrt{\frac{p}{3}} < k < q + \frac{2p}{3} \sqrt{\frac{p}{3}}$ (4)



(6) a) $P(x) = x^4 + ax^2 + bx$
 $x^2 + 1 = (x+i)(x-i)$
 $\therefore P(i) = i^4 + ai^2 + bi$
 $= 1 - a + bi$
 but $P(i) = i + 2$
 $\therefore 1 - a = 2$
 $bi = i$
 $\therefore a = -1 \quad b = 1$

(3)

b) if $P(x) = ax^4 + 4bx + c = 0$
 has a double root then
 it also a root of $P'(x) = 0$
 i.e. $P'(x) = 4ax^3 + 4b = 0$
 $\therefore x^3 = -\frac{b}{a}$

$\therefore x = \sqrt[3]{-\frac{b}{a}}$
 Subst this into $P(x)$
 $\therefore a \left(-\frac{b}{a}\right)^{\frac{4}{3}} + 4b \left(-\frac{b}{a}\right)^{\frac{1}{3}} + c = 0$
 $\therefore -\frac{ab}{a} \left(-\frac{b}{a}\right)^{\frac{1}{3}} + 4b \left(-\frac{b}{a}\right)^{\frac{1}{3}} = -c$
 $\therefore 3b \left(-\frac{b}{a}\right)^{\frac{1}{3}} = -c$
 cubing both sides
 $27b^3 \times -\frac{b}{a} = -c^3$
 $\therefore 27b^4 = ac^3$ (4)

c) $\alpha + \beta + \gamma = -\frac{b}{a} = 0$
 $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = p$
 $\alpha\beta\gamma = -\frac{d}{a} = -q$
 i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 0 - 2p$
 $= -2p$
 ii) $(\alpha - 1)(\beta - 1)(\gamma - 1) = \alpha\beta\gamma - \alpha\beta - \alpha\gamma - \beta\gamma + \alpha + \beta + \gamma - 1$
 $= -q - p - 1$

6c cont

$$\begin{aligned} \text{iii)} \quad x^3 &= -px - q \\ \alpha^3 &= -p\alpha - q \\ \beta^3 &= -p\beta - q \\ \gamma^3 &= -p\gamma - q \\ \therefore \alpha^3 + \beta^3 + \gamma^3 &= -p(\alpha + \beta + \gamma) - 3q \\ &= -3q \quad \text{---} \end{aligned}$$

$$\begin{aligned} \text{iv)} \quad x^4 &= -px^2 - qx \\ \alpha^4 &= -p\alpha^2 - q\alpha \\ \beta^4 &= -p\beta^2 - q\beta \\ \gamma^4 &= -p\gamma^2 - q\gamma \\ \therefore \alpha^4 + \beta^4 + \gamma^4 &= -p(\alpha^2 + \beta^2 + \gamma^2) - q(\alpha + \beta + \gamma) \\ &= -p \cdot 3p + 0 \\ &= -3p^2 \quad \text{---} \end{aligned}$$

⑦ a) $ma = mg - \frac{mgv}{T}$
 $\therefore a = g(1 - \frac{v}{T})$
 terminal velocity when $a = 0$
 i.e. $\frac{v}{T} = 1$

$$\therefore v = T$$

$\therefore T$ is terminal velocity. *

when projected upwards
 with initial velocity u
 i.e. at $t=0, x=0, v=u$

$$\begin{aligned} ma &= -mg - \frac{mgv}{T} \\ a &= -g \left(\frac{T+v}{T} \right) \end{aligned}$$

$$\therefore v \frac{dv}{du} = -g \left(\frac{T+v}{T} \right)$$

$$\therefore \left(\frac{Tv}{T+v} \right) \frac{dv}{du} = -g$$

$$\therefore T \int \frac{v}{T+v} dv = \int -g du$$

$$T \int \frac{v+T-T}{T+v} dv = -gx + C$$

$$T(v - T \ln(T+v)) = -gx + C$$

when $v=u, x=0$

$$\therefore C = T(u - T \ln(T+u))$$

$$\therefore x = T \left(\frac{u - T \ln(T+u) - v + T \ln(T+v)}{g} \right)$$

at max height $v=0$
 $\therefore H = T \left(\frac{u - T \ln(T+u) + T \ln T}{g} \right)$

$$= \frac{T}{g} \left(u + T \ln \left(\frac{T}{u+T} \right) \right)$$

$$H = \frac{uT}{g} + \frac{T^2}{g} \ln \left(\frac{T}{u+T} \right)$$

⑦

7b) $P(x) = 2x^3 - x^2 - 37x + 36 = 0$

$$P(1) = 2 - 1 - 37 + 36 = 0$$

1 is a root

by division other roots are $-\frac{9}{2}, 4$

for $P(x) = 2x^2 - 7x + k$

$$P(1), P\left(\frac{9}{2}\right), P(4) = 0$$

$$P(1) = 2 - 7 + k = 0 \therefore k = 5$$

$$P(4) = 32 - 28 + k = 0 \therefore k = -4$$

$$P\left(\frac{9}{2}\right) = \frac{81}{2} + \frac{63}{2} + k = 0 \therefore k = -72$$

7c) $(1+x)^n \left(1 + \frac{1}{x}\right)^n = \left((1+x) \left(1 + \frac{1}{x}\right) \right)^n$
 $= \left(1 + \frac{1}{x} + x + 1 \right)^n$
 $= \left(x + 2 + \frac{1}{x} \right)^n$
 $= \left(\left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 \right)^n$
 $= \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^{2n}$

$$(1+x)^n \left(1 + \frac{1}{x}\right)^n = \sum_{r=0}^n \binom{n}{r} x^r \sum_{r=0}^n \binom{n}{r} x^{-r} = \sum_{r=0}^n \binom{n}{r}^2$$

constant term is $\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \dots + \binom{n}{n}^2$
 $= \sum_{r=0}^n \binom{n}{r}^2$

$$\left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^{2n} = \sum_{r=0}^{2n} \binom{2n}{r} x^{\frac{2n-r}{2}}$$

for constant term $\frac{2n-r}{2} = 0$
 $n = r$

i.e. $2n \binom{n}{n}$

$$\therefore \sum_{r=0}^n \binom{n}{r}^2 = 2^n \binom{n}{n}$$

8) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$
 $\therefore \cos 5\theta = \text{Re}(\cos \theta + i \sin \theta)^5$

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta + 10 \cos^3 \theta \sin^2 \theta i^2 + 5 \cos \theta \sin^4 \theta i^4 \\ &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

$$16x^5 - 20x^3 + 5x = 0$$

let $x = \cos \theta$

roots of $\cos 5\theta = 0$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

$$\therefore x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{5\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10}$$

$$x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, 0, -\cos \frac{3\pi}{10}, -\cos \frac{\pi}{10}$$

solving algebraically

$$16x^5 - 20x^3 + 5x = 0$$

$$x(16x^4 - 20x^2 + 5) = 0$$

$$\therefore x = 0 \text{ or } x^2 = \frac{20 \pm \sqrt{400 - 320}}{16} = \frac{5 \pm \sqrt{5}}{8}$$

$$\therefore x = 0, \pm \sqrt{\frac{5 \pm \sqrt{5}}{8}}$$

now $\cos \frac{\pi}{10}$

$$0 < \cos \frac{3\pi}{10} < \cos \frac{\pi}{10}$$

ii) $\cos \frac{3\pi}{10} = \sqrt{\frac{5 - \sqrt{5}}{8}}$

iii) $\cos \frac{\pi}{10} = \sqrt{\frac{5 + \sqrt{5}}{8}}$

⑨

$$(8) b) I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$$

$$\text{let } u = x^n \quad v' = \sin x$$

$$u' = nx^{n-1} \quad v = -\cos x$$

$$I_n = \left[-x^n \cos x \right]_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx$$

$$= 0 + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx$$

$$\text{let } u = x^{n-1} \quad v' = \cos x$$

$$u' = (n-1)x^{n-2} \quad v = \sin x$$

$$\therefore I_n = n \left[x^{n-1} \sin x \right]_0^{\frac{\pi}{2}} - n(n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx$$

$$I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

$$\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$$

$$= I_4$$

$$I_4 = 4 \left(\frac{\pi}{2} \right)^3 - 4 \cdot 3 I_2$$

$$I_2 = 2 \left(\frac{\pi}{2} \right)^2 - 2 I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} \sin x \, dx = 1$$

$$\therefore I_2 = \pi - 2$$

$$I_4 = 4 \left(\frac{\pi}{2} \right)^3 - 12(\pi - 2)$$

$$= \frac{\pi^3}{2} - 12\pi + 24$$

(4)

(8) c)

in $\triangle ABD$

$$\cos \theta = \frac{m^2 + \frac{a^2}{4} - a^2}{am}$$

in $\triangle ACD$

$$\cos(\pi - \theta) = \frac{m^2 + \frac{a^2}{4} - b^2}{am} = -\cos \theta$$

$$\therefore m^2 + \frac{a^2}{4} - a^2 + m^2 + \frac{a^2}{4} - b^2 = 0$$

$$2m^2$$

8c) in $\triangle ABD$

$$\cos \theta = \frac{m^2 + \frac{a^2}{4} - c^2}{2 \cdot \frac{a}{2} \cdot m} = \frac{4m^2 + a^2 - 4c^2}{4am}$$

in $\triangle ACD$

$$-\cos \theta = \frac{m^2 + \frac{a^2}{4} - b^2}{2 \cdot \frac{a}{2} \cdot m} = \frac{4m^2 + a^2 - 4b^2}{4am}$$

$$\therefore \frac{4m^2 + a^2 - 4c^2}{4am} + \frac{4m^2 + a^2 - 4b^2}{4am} = 0$$

$$8m^2 = 4b^2 + 4c^2 - 2a^2$$

$$m = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} \quad (3)$$

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